

Suhani Surya

Toronto, ON | Open to relocation

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EDUCATION

University of Waterloo

Expected April 2029

Candidate for Systems Design Engineering (Co-op) - BASc

Waterloo, ON

TECHNICAL SKILLS & INTERESTS

Programming Languages and Frameworks: Java, Python, JavaScript, HTML, CSS, C++, React, Angular, Flask

Databases: PostgreSQL, MySQL

Developer Tools: Postman, Git, GitHub, G-Suite, Office 365, PowerPoint, Excel, Notion, Asana, Jira, Excel

VBA/Macros, MatLab, ProjectWise

Hardware Skills: Circuitry, Microcontroller Programming (Raspberry Pi), Soldering, Sensor Integration

Design Skills: AutoCAD, Solidworks, Blender, Figma, Canva

Interests and Hobbies: Photography, Pilates, Vintage Shopping, Badminton, Tennis, Travelling

EXPERIENCE

Software Engineering Intern

Sept 2025 – Present

SS&C Technologies

Toronto, ON

- Built enterprise-grade web applications for SS&C's Lyric platform serving **100+ global financial institutions**, implementing user-facing features with Angular and TypeScript.
- Collaborated with cross-functional agile teams using Jira and Confluence to deliver production-ready microservices for SS&C's Lyric recordkeeping platform, supporting real-time operations for **18,000+ financial and healthcare organizations** with Git-based version control workflows.
- Executed comprehensive API testing strategies using Postman, enhanced Postgres database queries and schema design, and implemented CI/CD best practices including automated testing and deployment pipelines.

Rail Systems Engineering Intern

Jan 2025 – Apr 2025

Hatch

Mississauga, ON

- Engineered a C++ TrainOps algorithm using OOP, arrays, and control flow to simulate **16 train routes** and detect collisions, reducing manual testing by **30%**.
- Reviewed and updated 250+ CAD signalling/as-built drawings with engineers to ensure utility compliance.
- Deployed an EVE-NG virtual lab simulating 10+ IoT devices, saving **20+ hours** in hardware testing for cybersecurity threat analysis.

Software Developer

Sept 2023 – Jan 2024

First Robotics - Team 3161

Oakville, ON

- Implemented proportional-integral-derivative (PID) to improve the robot's arm mobility by **15%**.
- Created **Java routines** automating **5+ robot movement tasks**, reducing manual intervention by **20%**.
- Created documentation and reports for **2 software modules**, facilitating future maintenance.

PROJECTS

Cloakify

Sept 2025

Personal Project | **Tech Stack:** JavaScript, HTML/CSS, Flask, Vercel

- Built an interactive web app that simulates an **invisibility cloak effect** by processing real-time video streams with optimized browser-based computer vision, achieving smooth frame rates and sub 100ms latency.
- Engineered the system to run fully on the **client side**, leveraging JavaScript and HTML5 Canvas for video processing, ensuring smooth performance without heavy server load.
- Deployed the project on **Vercel** for global accessibility with a streamlined CI/CD pipeline from GitHub branches.

[Github Repo](#)

Traffic Light Simulator

Jun 2024

Personal Project | **Tech Stack:** Raspberry Pi Pico and Micro Python

- Configured a **traffic light simulation** on Raspberry Pi Pico, integrating Micro Python to control three LED signals for accurate signal transitions.
- Programmed and optimized GPIO pin configurations, achieving **precise millisecond-level control** over electronic components to **simulate real-world traffic sequences** with adjustable timing and audio signals.
- Programmed a 100+ line MicroPython script to automate light transitions, leveraging efficient sleep functions to maintain cycle timing with **sub-10ms** accuracy.

[Github Repo](#)